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MedSentry: AI-Powered Medication Safety & Adherence

FULL STACK DEVELOPMENT

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Domain Introduction

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APPLICATION DOMAIN

Healthcare technology + medication management.

MedSentry connects medicine identification, prescription review, reminders, and family-based care coordination in one mobile workflow.

REAL-WORLD CONTEXT

- Medicine labels and prescriptions can be hard to interpret.
- Daily regimens require timely dose tracking.
- Caregivers may coordinate medicines for dependants.
- Antibiotic misuse is a public-health concern.

WHY FULL STACK?

- Mobile-first user experience
- API orchestration and AI routing
- Relational data + access controls
- OCR and medicine matching
- Reminders, family collaboration, and analytics

Purpose: improve medicine understanding and adherence support; the app does not replace professional medical advice.

Problem Formulation

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CORE PROBLEM

People must interpret medicine information, remember dose timing, track treatment, and coordinate care—often across separate tools.

The project aims to bring these tasks into a connected workflow.

LIMITATIONS TO ADDRESS

- Manual prescription reading is error-prone.
- Basic reminders do not explain medicine context.
- Missed doses and early stops can go unnoticed.
- Family sharing needs explicit permissions.

INTENDED USERS

Patients managing daily medicines

Family members and caregivers

Users scanning prescriptions or medicine packs

Administrators reviewing usage and aggregated AMR trends

Problem statement: build a secure, AI-assisted full-stack app that unifies medication information, dose adherence, and family support.

Clinical recommendations must remain guarded, transparent, and subject to pharmacist / clinician confirmation.

Motivation and Objectives

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MOTIVATION

Make medicine management simpler and more connected.

Use AI conversation as an accessible starting point, with scanning, scheduling, and caregiver support behind it.

Encourage safer, more responsible antibiotic use.

1

01 Scan & extract

Use camera + OCR to structure medicine and prescription details.

2

02 Assist through AI

Provide a chat-first Consultant for medication questions and image-based input.

3

03 Track adherence

Store medication courses and dose outcomes; schedule reminders.

4

04 Promote stewardship

Surface drug information, interaction checks, and antibiotic awareness.

5

05 Coordinate family care

Apply role-based access and support caregiver alerts and admin analytics.

The objectives describe the intended product scope; evaluation must be reported separately from UI design evidence.

Proposed System

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1. CAPTURE

Scan a medicine pack or prescription image.

2. RECOGNIZE

ML Kit OCR + line clustering and text cleanup.

3. UNDERSTAND

Match drug references; route questions to AI with safety checks.

4. TRACK

Confirm details, create a course, and log taken / missed doses.

5. COORDINATE

Send reminders, caregiver alerts, and provide aggregate insights.

AI-FIRST INTERACTION

The Chatbot / AI Consultant is the primary conversational interface: the user asks a question or attaches an image, the frontend sends a protected request to the backend router, safety checks are applied, and the response returns to the chat. Dedicated screens support tracking, family care, and administration.

Medicine matching and AI responses assist users; they are not a definitive diagnosis or substitute for a prescriber.

Full Stack Technologies Used

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LAYER	TECHNOLOGY / ROLE
Frontend	React Native + Expo + TypeScript; React Navigation / native stack
Primary interaction	AI Chatbot + AI Consultant screens; image and text input
Backend / API	Supabase Edge Functions (Deno): ai-router, admin-api, verify-email
Client / Data API	supabase-js; JWT session persistence via AsyncStorage; PostgREST + RPC
Database / security	PostgreSQL 15; Row Level Security (RLS), stored functions, triggers, pg_trgm
OCR / matching	Expo Camera + Google ML Kit OCR; spatial line clustering, Levenshtein + trigram matching
AI providers	Gemini 1.5 Flash (primary); Groq Llama 3.3 70B (fallback); OpenRouter where configured
Services	Supabase Auth / Storage / Realtime; Expo Notifications; email via Resend

The AI routing options reflect the supplied architecture and code-asset notes; actual provider selection depends on deployed configuration and secrets.

System Architecture

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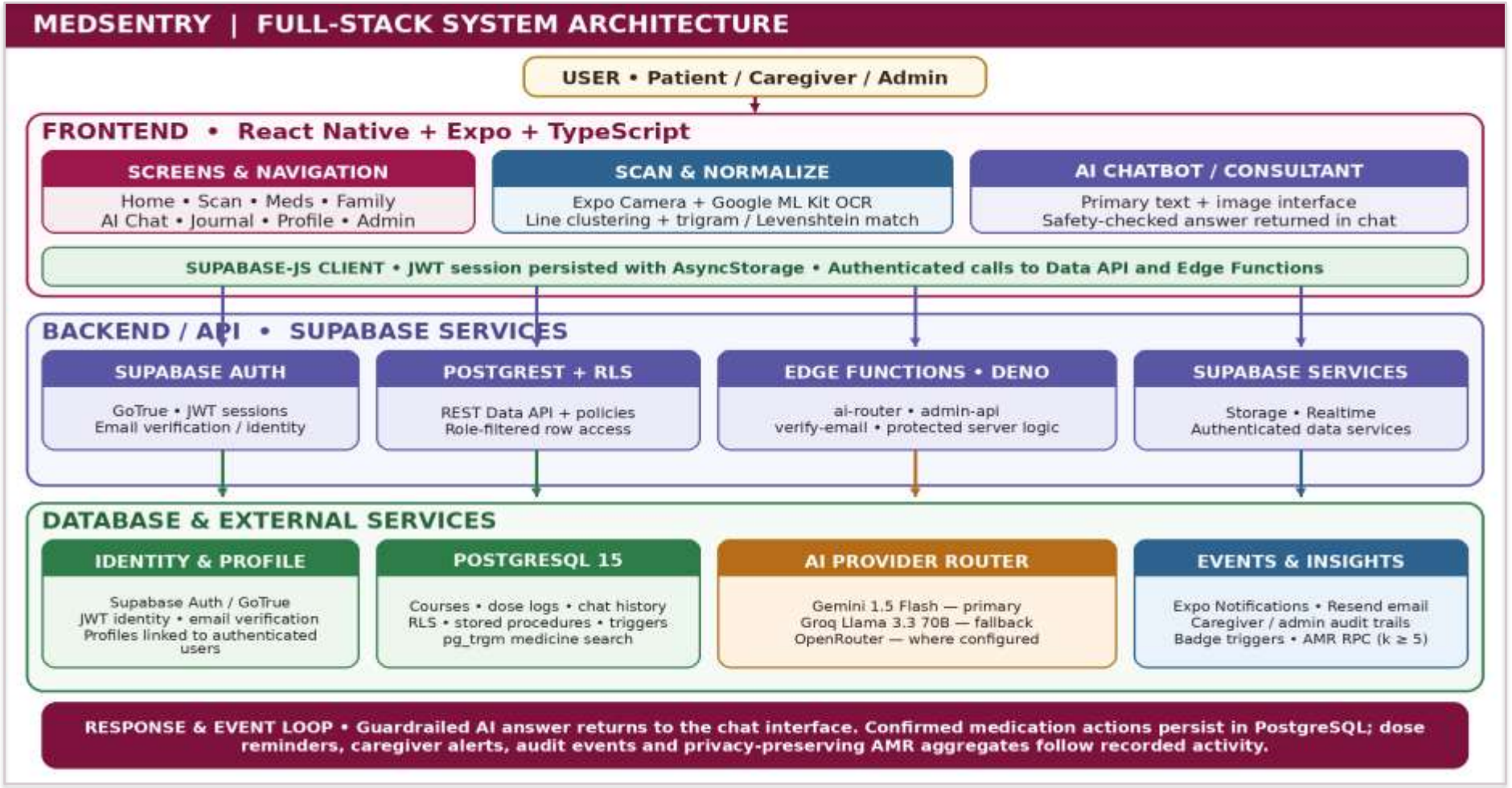
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Security boundary: provider secrets remain server-side; database reads/writes are governed by authentication and RLS.

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Datasets Used & Their Role

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MedSentry combines medicine-reference data with user-entered medication records and privacy-protected aggregate insights.

1 MEDICINE FORMULARY

- Verified medicine formulary (documented as 10,000+ entries)
- Fuzzy matching helps resolve OCR spelling errors
- Matched details are shown for user review

2 WHO AWaRe CLASSIFICATION

- WHO AWaRe: Access, Watch, Reserve
- Adds antibiotic-stewardship context to medicine information
- Supports awareness—not prescribing decisions

3 APP / OPERATIONAL DATA

- Profiles, medication courses and dose logs
- Chatbot messages and family-care records
- Stored in PostgreSQL with authentication + RLS

HOW THE DATA IS USED IN THE APPLICATION

SCAN
Camera image → ML Kit OCR

→

MATCH
Line clustering + Levenshtein / trigram search

→

VERIFY
User reviews extracted medicine fields

→

TRACK
Confirmed courses and dose events are stored

→

AGGREGATE
AMR trends only after privacy thresholds

Scope note: supplied materials do not identify exact dataset filenames, versions, or licenses; no model-training dataset is documented.

Chatbot / AI Agent Interaction

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AI CHAT — PROVIDED UI CAPTURE



1. ASK → Text or image

2. ROUTE → Edge Function

3. CHECK → Allergies + drug data

4. REPLY → Answer + usage log

QUERY 1 • MEDICINE INFORMATION

“What should I check on my medicine strip?”

Response pattern: Check the name, strength, expiry, batch details, and prescribed instructions. Ask a pharmacist if unclear.

QUERY 2 • MISSED DOSE

“I missed my afternoon dose. What now?”

Response pattern: Follow the medicine-specific instructions; do not double doses. Ask a pharmacist or prescriber if unsure.

QUERY 3 • ANTIBIOTIC COURSE

“I feel better. Can I stop early?”

Response pattern: Do not stop or change a prescribed course without contacting the prescribing clinician.

Safety guardrails: no unsupported diagnosis or dose changes; serious symptoms and medication uncertainty should be escalated to a qualified professional.

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Implementation and Key Modules

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Frontend	OCR	AI Router	Data & Security	Care & Insights
React Native / Expo	ML Kit + matching	Supabase Edge Functions	Postgres + RLS	Family, reminders, AMR

OCR INPUT



OCR OUTPUT

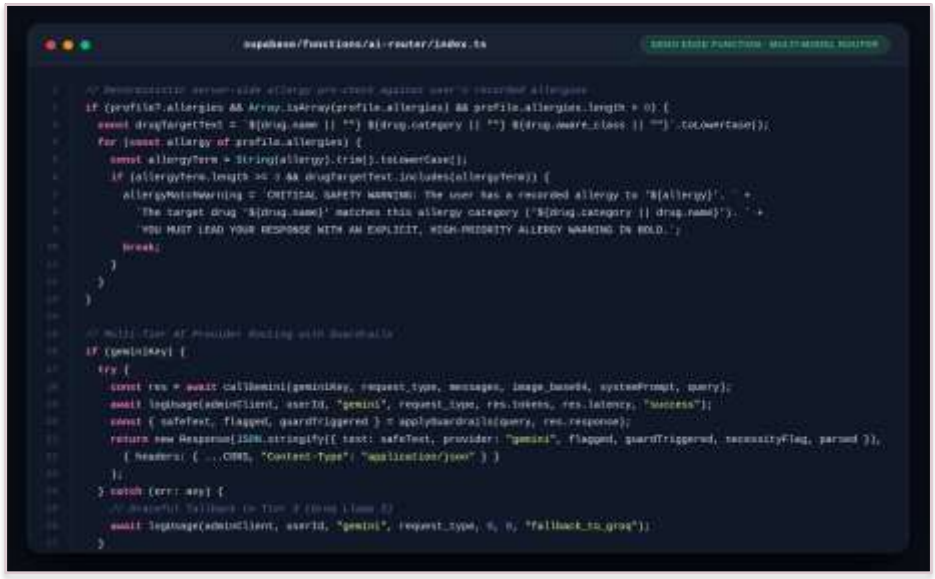


AI CHAT UI



OCR flow: capture → recognize → review extracted fields → confirm before creating a medication course.

BACKEND AI ROUTER — CODE SCREENSHOT



Backend focus: server-side provider routing, protected admin calls, database policies, and usage logging.

Images are selected from report_assets/app_screenshots and report_assets/code_snippets; no pre-made diagram images from report_assets/diagrams are used.

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Results and Discussion

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HOME DASHBOARD



Dose cards, progress, next-dose reminders.

AI CHAT



User prompts and chat-style guidance.

AMR HEATMAP



Aggregated surveillance-style interface.

EVIDENCE IN THE PROVIDED ASSETS

- Mobile screens cover scanning, medicines, AI chat, family, profile, and admin.
- Code screenshots illustrate AI routing and OCR-related modules.

VALIDATION STILL NEEDED

No repeatable end-to-end test report or measured OCR accuracy, response latency, or adherence improvement was included. Do not claim these as measured results.

Next demo: run scan → chat → confirm → log dose live.

The screenshots are UI evidence from the supplied asset bundle; they are not a substitute for recording a live end-to-end demo.

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Conclusion and Future Enhancement

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CONCLUSION / CONTRIBUTION

MedSentry brings together:

- Chat-first medication assistance
- OCR-supported medicine and prescription capture
- Medication courses, dose logs, and reminders
- Family roles and caregiver support
- Admin visibility and AMR-oriented aggregate insights

FUTURE ENHANCEMENTS

- Test OCR and fuzzy matching against labeled prescriptions.
- Validate medication / interaction data with trusted sources.
- Benchmark provider fallback, latency, and failure recovery.
- Expand multilingual and accessibility support.
- Improve clinical review, privacy checks, and deploy-time monitoring.

Safety boundary: AI guidance is informational and must not replace qualified medical advice.

GitHub Repository

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GITHUB https://github.com/Subhash-777/MedSentry

REPOSITORY STRUCTURE

mobile-app/ — React Native + Expo client

supabase/functions/ — Edge APIs and AI routing

supabase/migrations/ — schema, RLS, RPCs, triggers

datasets/ + scripts/ — reference data and seeding utilities

docs/plan.md — implementation plan / architecture

README.md — repository entry point

IMPORTANT MODULES

- OCR: spatial clustering + medicine matching
- AI router: provider fallback + guardrails
- Dose logging and reminder flows
- Caregiver access / RLS and audit trail
- Admin metrics and AMR aggregation

DOCUMENTATION

Use docs/plan.md for detailed technical design. Expand README with setup, environment variables, database migrations, and run instructions.

Before submission: ensure the repository has reproducible setup steps, non-secret environment examples, migrations, and a recorded live demo.

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